

Labsheet 4



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LCD



Name:

Student ID:

Section (Day/Time):

COMPUTER NAME:

UNIVERSITY OF JORDAN
SCHOOL OF ENGINEERING
DEPARTMENT OF COMPUTER ENGINEERING
EMBEDDED SYSTEMS LABORATORY CPE0907334
Labsheet 4: LCD

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(Pre-lab) Part 1: Code Analysis and Understanding:

Answer the following questions regarding Code1:[LDC1.asm](#) of the experiment.

1. What changes to the instruction at line 57 are necessary in order for the cursor to be blinking?

2. What change(s) would you make to the code in order for it to start displaying the characters at approximately the middle of the visible screen (i.e. at address 6)

(Pre-lab) Part 2: Suppose that you will store your Arabic characters at CG RAM Locations 0, Location 1, Location 2 and location 3. Complete the modified main subroutine down with the appropriate values in the Movlw instructions.

Main	Movlw _____ ;Display character stored in location 00 which in
	Call send_char ;this case is your first charcter in CG-RAM
	Movlw _____
	Call send_char
	Movlw _____
	Call send_char
	Movlw _____
	Call send_char
	Goto Main ; This loop makes the character rotate continuously

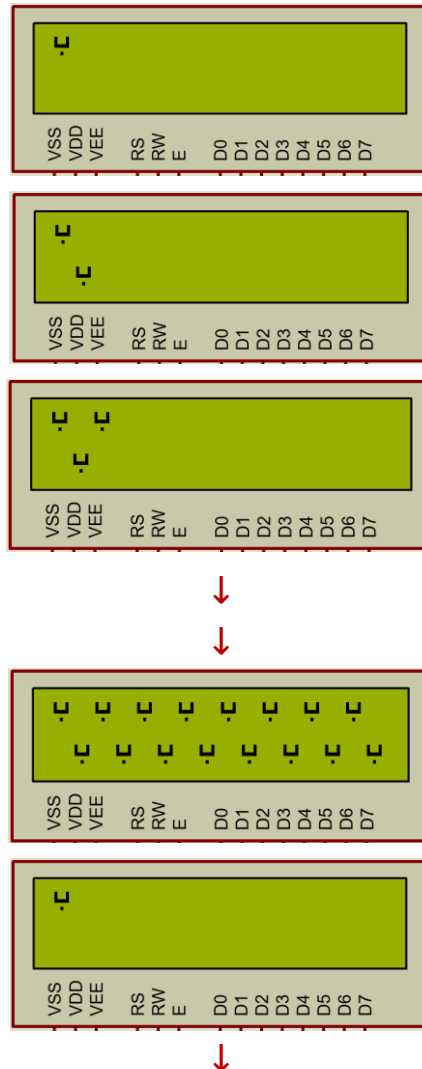
(Pre-lab) Part 3: On the grid given below, shade the squares required to generate the first letter of your name in Arabic. If the first letter of your name is (ا), generate the character for the second letter in your name. Then, write the instructions needed to store this character in CG(1).

Draw your character below						Replace each shaded cell with one and not shaded ones with zero.									Data in Hex
B4	B3	B2	B1	B0	→	B7	B6	B5	B4	B3	B2	B1	B0		
						0	0	0							0x
						0	0	0							0x
						0	0	0							0x
						0	0	0							0x
						0	0	0							0x
						0	0	0							0x
						0	0	0							0x
						0	0	0							0x
						0	0	0							0x
					0	0	0							0x	

; Code to store the first letter of your name in Arabic

Part 3: Code Modification

In this part, you are required to modify the code in [LCD2 - Animating Stickman.asm](#) to display the first Arabic letter of your name in a zigzag fashion as shown in the figure below. When the last position is reached on the second row, the screen is cleared and the operation is repeated. You can use the Embedded_Ex4 Proteus file to test your code. Assume that you are using 16x2 display.



Hints:

1. You will need to use the **Set Display Address** command to select the display location. Check Figure 2 in the tutorial to know the addresses of the locations to be used.
2. For proper display of the pattern, it is advised to turn off the cursor when you initialize the LCD in the Init subroutine.
3. To reduce the program size, you may write a lookup table that stores the display addresses and call this table in a loop.

Ask your engineer to check the run.